- Human beings reproduce sexually and are viviparous.
- In humans, the reproductive phase starts after puberty.
- It involves:
  - Gametogenesis
  - Insemination
  - Fertilisation
  - Implantation
  - Gestation
  - Parturition

The Male Reproductive System

- It is located in the pelvic region.
- It consists of:
  - A pair of testes
  - Accessory glands and ducts
  - External genitalia

Testes

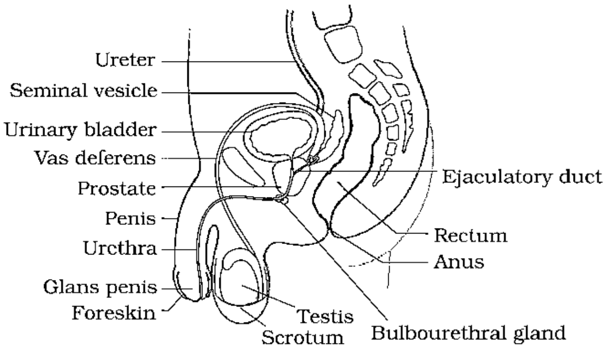

Figure: Diagrammatic sectional view of male pelvis showing reproductive system

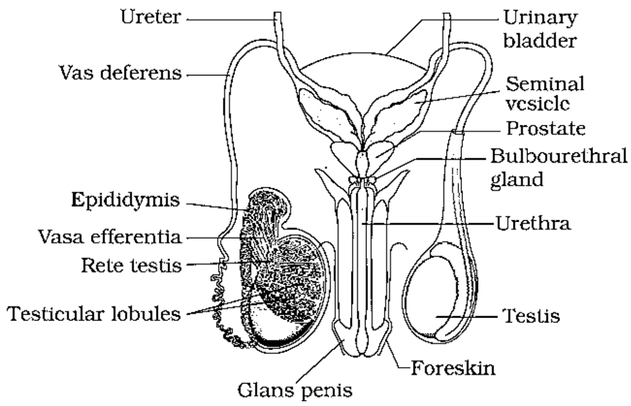

Figure: Diagrammatic view of male reproductive system (parts of testis is open to show inner details)

- Situated within the **scrotum**, which protects the testes and also helps in maintaining the temperature (2-2.5^{0}C lower than the normal internal body temperature)
- Each testis is 4 to 5 cm in length, and 2 to 3 cm in width, and has about 250 compartments called **testicular lobules**.
- Testicular lobules have **seminiferous tubules** which are the sites of sperm formation.
- Seminiferous tubules are lined by two types of cells:
  - **Male germ cells** – they undergo meiosis to form sperms.
  - **Sertoli cells** – they provide nourishment to the germ cells.
- Region outside the seminiferous tubules is called the interstitial space, which contains **Leydig cells** (interstitial cells). The Leydig cells produce male sex hormone - **androgens (testosterone).**

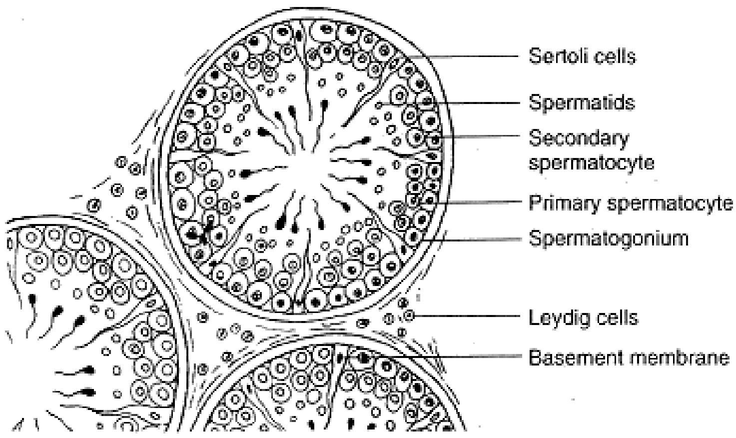

Figure: Diagrammatic sectional view of seminiferous tubule

Accessory ducts

- It include:
  - Rete testis
  - Vasa efferentia
  - Epididymis
  - Vas deferens

- The seminiferous tubules open into the **vasa efferentia** through the **rete testis**.
- The vasa efferentia open into the **epididymis** (sperms undergo physiological mutaration, acquiring increased motility and fertilizing capacity) which leads to the **vas deferens**. The vas deferens opens into the urethra along with a duct from the seminal vesicle called the **ejaculatory duct.**
- The ejaculatory duct stores the sperms and transports them to the outside
- The urethra starts from the urinary bladder, extends through the penis and opens via the **urethral meatus**.

Accessory glands

- It include:
  - A pair of seminal vesicles
  - Prostate gland
  - A pair of bulbourethral glands or Cowper's glands
- The secretions of these glands make up the seminal plasma (pH 7.4 and forms 60% of the volume of semen) and provide nutrition which is **rich in fructose, calcium and certain enzymes** and a medium of motility to the sperms.

External genitalia

- The penis is the male external genitalia.
- It is made up of special tissue (corpora carvernosa and corpora spongiosum) that helps in erection of the penis to facilitate insemination.
- The enlarged end of penis called the **glans penis** is covered by a loose fold of skin called **foreskin** (prepuce).

The Female Reproductive System

- It is located in the pelvic region:
- It includes:
  - A pair of ovaries
  - A pair of oviducts
  - Uterus
  - Cervix
  - Vagina
  - External genitalia
  - Mammary glands (not part of the reproductive system, but aids in child care)

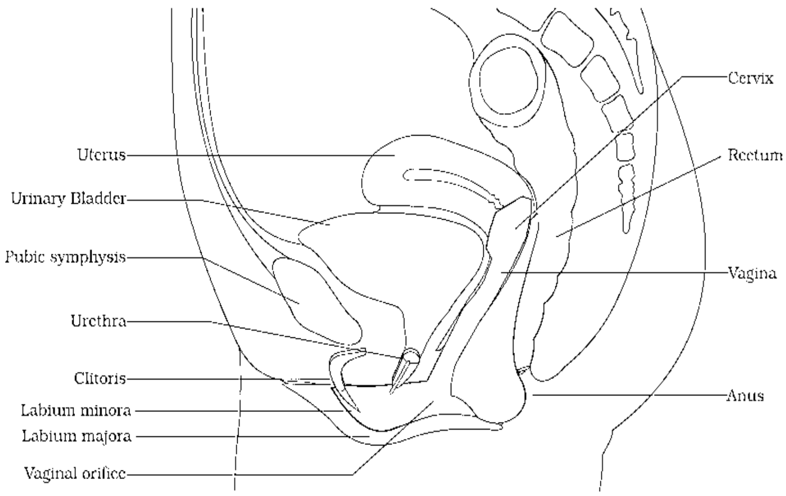

Figure: Diagrammatic sectional view of female pelvis showing reproductive system

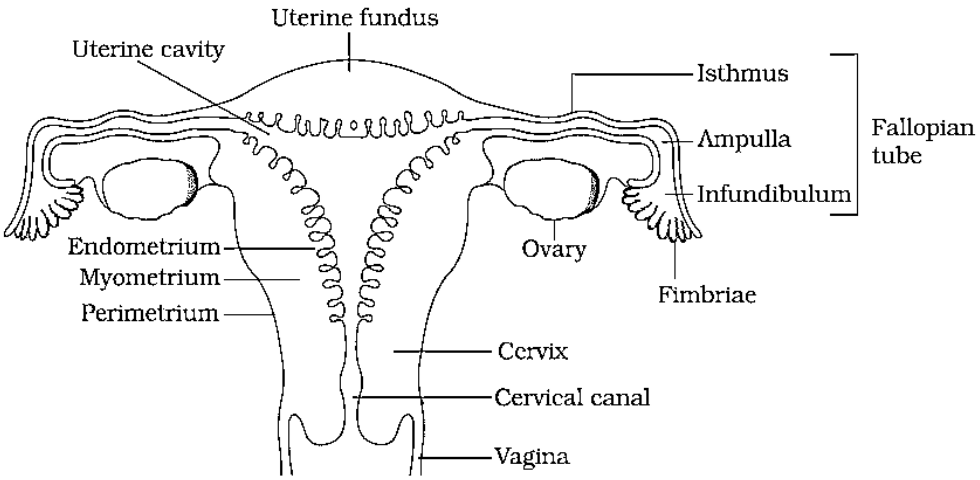

Figure: Diagrammatic sectional view of the female reproductive system

Ovaries

- They are the primary female sex organs. They produce the ovum and other ovarian hormones.
- They are located in the lower abdomen, and are 2 to 4 cm in length.
- They are connected by ligaments to the pelvic walls and to the uterus.
- Each ovary is covered by epithelium, and contains the ovarian stroma.
- The ovarian stroma is made up of:
  - Peripheral cortex
  - Inner medulla

Oviducts

- They are also called **fallopian tubes**.
- They are 10 to 12 cm long, and extend from the ovary to the uterus.
- The part of each oviduct lying towards the ovary is funnel shaped, and is called **infundibulum**. It has finger-like projections called **fimbriae**.
- The infundibulum leads to the ampulla, and then to the isthmus, which has a narrow lumen opening into the uterus.

Uterus

- It is also called **womb**, and is **pear shaped**.
- It is connected to the pelvic walls by ligaments.
- The uterine wall consists of:
  - External perimetrium
  - Middle myometrium
  - Internal endometrium, which lines the uterine cavity
- The endometrium undergoes changes during the menstrual cycle.

Cervix and Vagina

- The cervix connects the uterus to the vagina.
- The cervix and the vagina constitute the **birth canal**.

External Genitalia

- Consists of:
  - Mons pubis – Fatty tissue covered by skin and pubic hair
  - Labia majora – Extends from mons pubis and surrounds the vaginal opening
  - Labia minora – Fold of skin beneath the labia majora
  - Hymen – is a thin membrane which partially covers the vaginal opening
  - Clitoris – is a erectile finger like structure which lies at the junction of labia minora

NOTE –

- The hymen is often torn during the first coitus (intercourse).
- However, it can also be broken by a sudden fall or jolt, insertion of a vaginal tampon, active participation in some sports like horseback riding, cycling, etc.
- In some women the hymen persists even after coitus.
- In fact, the presence or absence of hymen is not a reliable indicator of virginity or sexual experience.

Mammary Glands

- Mammary glands are a pair of rounded thoracic prominences.
- Mammary glands develops during puberty but becomes active only after child birth.
- Present in all female mammals
- It is **paired** and is **glandular**.
- Each breast contains 15 to 20 mammary lobes with **alveoli** (modified sweat glands) which secrete milk.
- The alveoli open into the **mammary tubules**, which unite to form a **mammary duct**.
- Many mammary ducts constitute the **mammary ampulla**, which is connected to the **lactiferous duct**.
- Secretion of milk is under the control of prolactin hormone of anterior pituitary. While the ejection of milk is under control of oxytocin hormone of posterior pituitary.

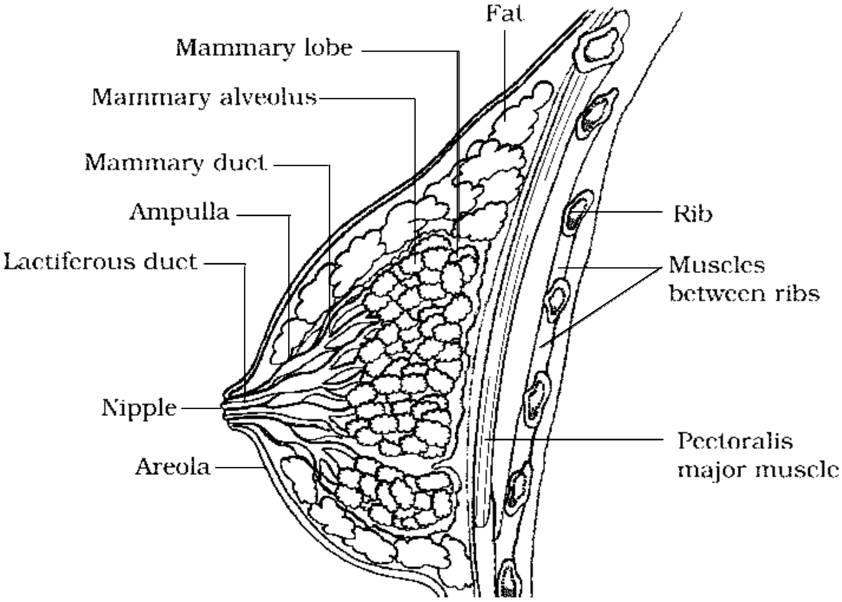

Figure: A Diagrammatic sectional view of Mammary gland

Gametogenesis

- The process of formation of gametes is called **gametogenesis.**
- The testis and ovary produce the male and female gametes respectively by gametogenesis (spermatogenesis in males and oogenesis in females).

Spermatogenesis

- The process of formation of sperms in testis is called **spermatogenesis.**
- In males, sperms are produced by the **spermatogonia** (immature germ cells), which are present in the inner walls of the seminiferous tubules.
- Spermatogonia increase in number by mitosis. These are diploid.
- Some of the spermatogonia called **primary spermatocytes** periodically undergo meiosis.
- After the first meiotic division, two haploid and equal **secondary spermatocytes** are formed.
- These further undergo meiosis to give rise to four haploid **spermatids.**
- These spermatids are converted into sperms by **spermiogenesis.**

- The sperm head gets embedded in the Sertoli cells after spermiogenesis and is released into the cavity of the seminiferous tubules and this process is called **Spermiation**.

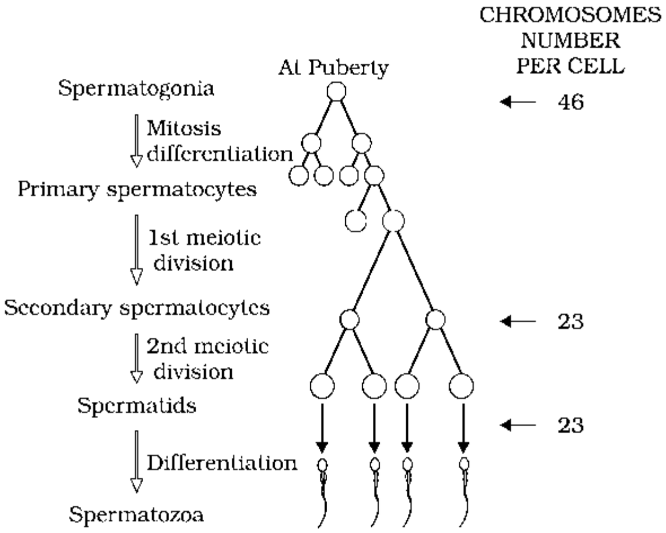

Figure: Schematic representation of Spermatogenesis

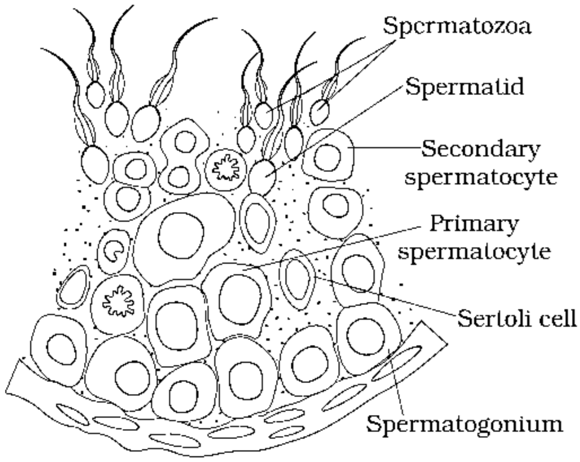

Figure: Diagrammatic sectional view of seminiferous tubule (enlarged)

Hormonal Regulation of Spermatogenesis

- Spermatogenesis starts at puberty by the action of the gonadotropin releasing hormone (GnRH), which in turn causes the release of two gonadotropins called Luteinizing Hormone (LH) or Interstitial Cell Stimulating Hormone (ICSH) and Follicle Stimulating Hormone (FSH).
- LH acts on Leydig cells and causes them to release androgens, which stimulate the process of spermatogenesis while the FSH acts on the Sertoli cells, which help in spermiogenesis.

Structure of a Sperm

- A mature sperm consists of:
  - Head
  - Neck
  - Middle piece
  - Tail

- The whole sperm is enclosed in a plasma membrane.
- The head consists of a haploid nucleus and a cap-like **acrosome**, which contains enzymes that aid in fertilisation.
- The middle piece contains several mitochondria, which produce energy for the motility of the sperm.
- Sperms released by the seminiferous tubules are transported by the accessory ducts.
- Secretions of epididymis, vas deferens, seminal vesicles, and prostate are essential for maturation and motility of sperms.

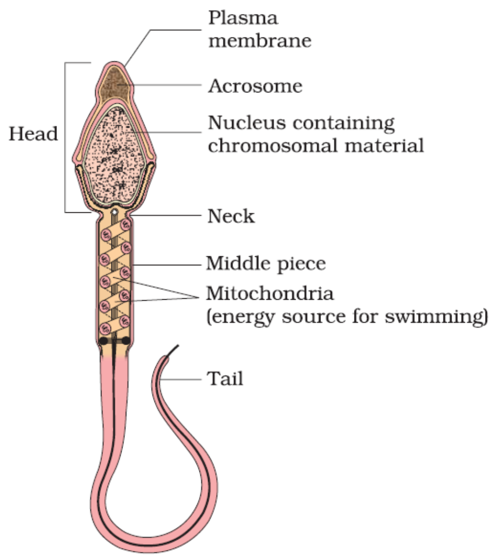

Figure: Structure of a sperm

Oogenesis

- The process of formation of ovum is called **oogenesis.**
- It starts during embryonic growth and millions of gamete mother cells (**oogonia**) are formed in the foetal ovary.
- These cells undergo meiosis, but get temporarily arrested at the prophase and are called **primary oocytes**.
- Each primary oocyte then gets surrounded by a layer of granulosa cells and is called the **primary follicle.**

- Before reaching puberty, a large number of primary follicles degenerate and the remaining (60,000 – 80,000) ones get surrounded by more layers of granulosa cells and new theca and are called **secondary follicles**.
- The secondary follicles are then converted into **tertiary follicles** that have characteristic fluid-filled cavity called **antrum.** At this stage, the primary oocyte present within the tertiary follicle completes meiosis, which results in the formation of haploid secondary oocyte and a tiny polar body.
- This tertiary follicle further changes into the **Graafian follicle**. The secondary oocyte is surrounded by the zone pellucida.
- Then the Graafian follicle ruptures to release the ovum by and the process is called **ovulation.**

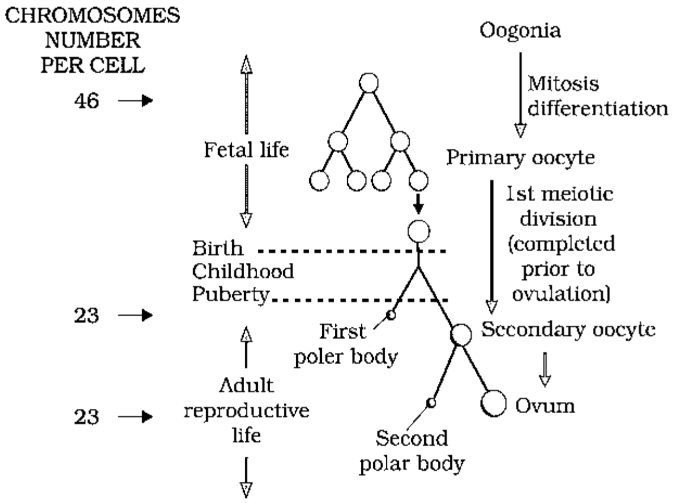

Figure: *Schematic representaiton of Oogenesis*

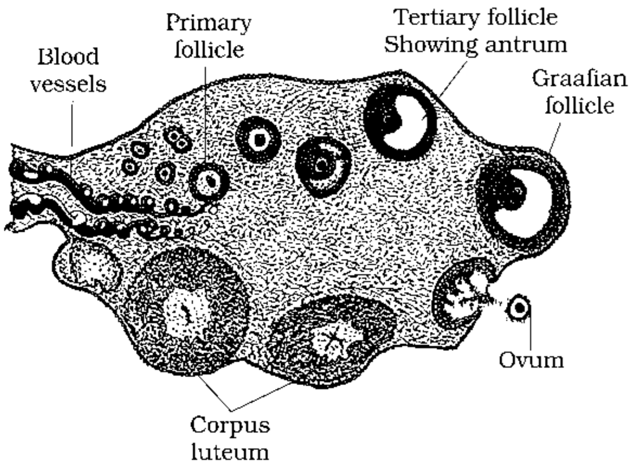

Figure: *Diagrammatic sectional view of ovary*

Menstrual Cycle

- The cycle of events starting from one menstruation till the next one is called the **menstrual cycle**.

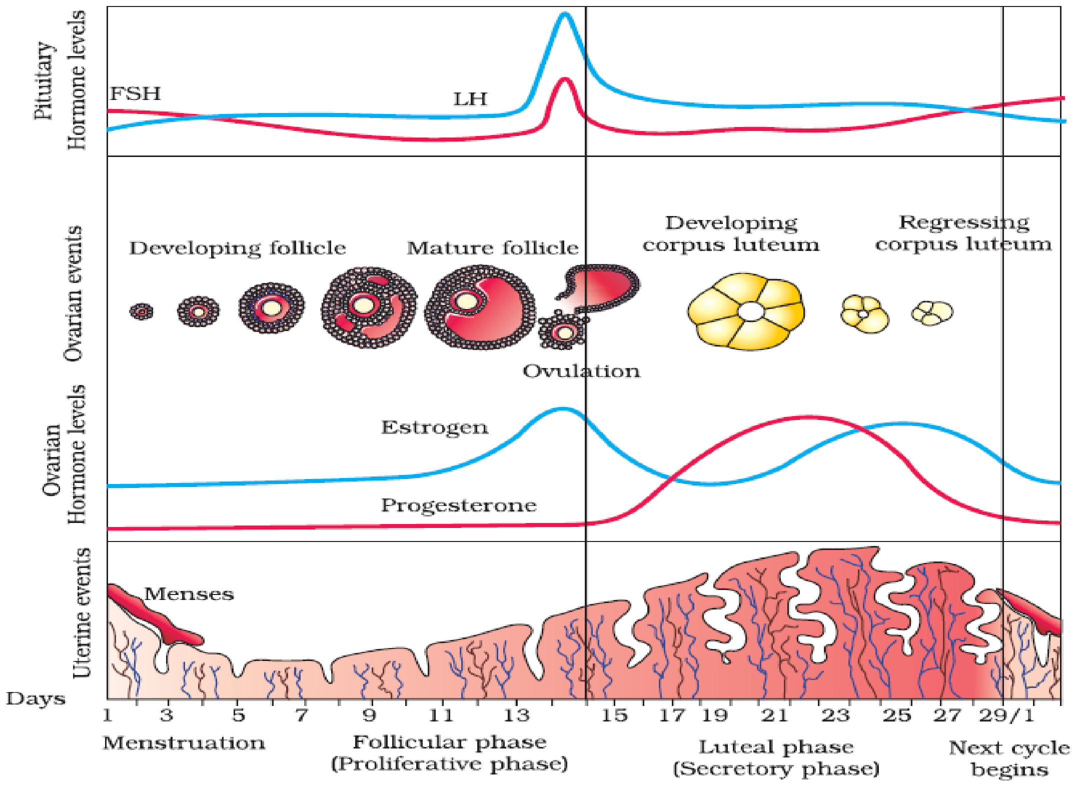

Figure: Diagrammatic representation of various events during a menstrual cycle

- Menstrual cycle is the reproductive cycle in all primates and begins at puberty (between the age of 12 – 15) and is called as **menarche.**
- In human females, menstruation occurs once in 28 to 29 days.
- During the middle of the menstrual cycle, one ovum is released.
- Menstrual cycle can be divided into 4 phases, namely Menstrual phase, Follicular phase (proliferative phase), ovulatory phase and Luteal phase (secretory phase).

1. **Menstrual phase -**The cycle starts with the **menstrual flow/menses/menstruation** (3 to 5 days). Menstruation is the periodic discharge of **blood**, **mucus** and **cellular debris** from the uterus through vagina but this occurs only when the ovum is not fertilised.

2. **Follicular phase (proliferative phase) -** In this phase, the primary follicles mature into the Graffian follicles and the regeneration of the endometrium. These changes are brought about by ovarian and pituitary hormones. In this phase, the release of gonadotropins (LH and FSH) increases. This causes follicular growth and the growing follicles produce oestrogen which causes the regeneration of endometrium.

3. **Ovulatory phase -** The LH and FSH are at their peak in the middle of the cycle (14^{th} day)**.**The rapid secretion of LH leading to the maximum level during the mid-cycle called **LH surge** and cause the rupture of the Graffian follicles to release ovum. The process of release of ovum from the ovary is called **Ovulation.**

4. **Luteal phase (secretory phase) -** The remains of the Graffian follicles get converted into the corpus luteum, which secretes progesterone for the maintenance of the endometrium. In the absence of fertilisation, the corpus luteum degenerates, thereby causing the disintegration of the endometrium and the start of a new cycle.

- In humans, the menstrual cycle ceases to operate at the age of 50 years. This phase is known as the **menopause**.

Fertilisation and Implantation

- During coitus, the semen is released into the vagina, passes through the cervix of the uterus and reaches the **ampullary-isthmic junction** of the fallopian tube.
- The ovum is also released into the junction for fertilisation to occur.
- The process of fusion of the sperm and the ovum is known as **fertilisation** or **syngamy**.

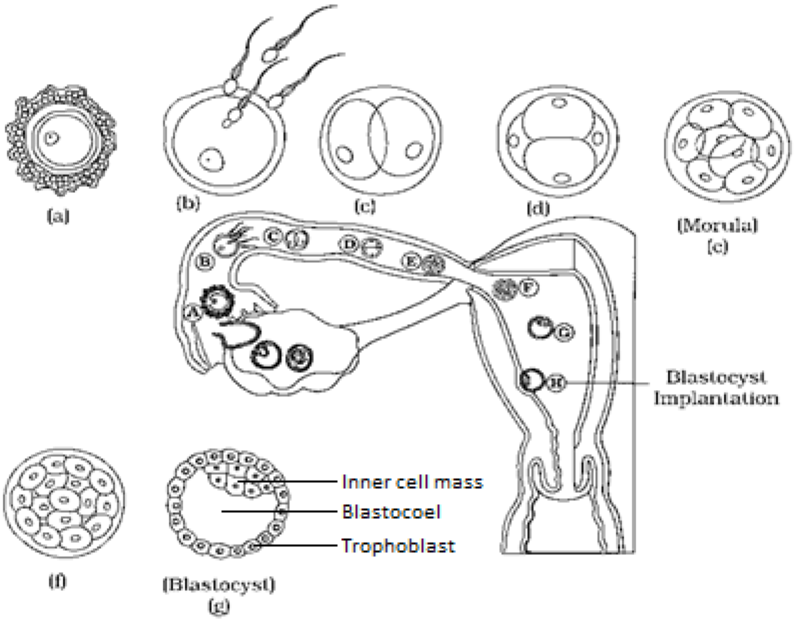

Figure: (a) Transport of ovum, (b) fertilisation, and (c) passage of growing embryo - 2 celled stage, (d) 4 celled stage, (e) 8 celled stage, (f) 16 celled stage embryo passing through fallopian tube

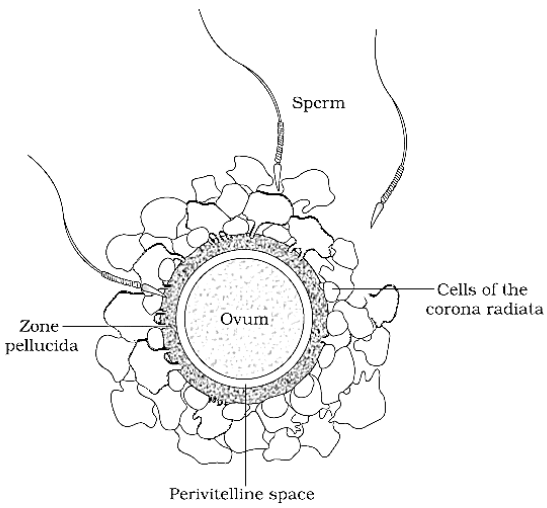

Figure: Ovum surrounded by few

- During fertilisation, the sperm induces changes in the **zona pellucida** and blocks the entry of other sperms. This ensures that only one sperm fertilises an ovum.
- The enzymatic secretions of the acrosomes help the sperm enter the cytoplasm of the ovum.
- This causes the completion of meiotic division of the secondary oocyte, resulting in the formation of a haploid **ovum** (ootid) and a secondary polar body.
- Then, the haploid sperm nucleus fuses with the haploid nucleus of the ovum to form a diploid **zygote**.
- The mitotic division starts as the zygote moves through the isthmus of the oviduct called **cleavage** towards the uterus and forms 2, 4, 8, 16 daughter cells called **blastomeres**.
- The 8−16 cell embryo is called a **morula**, which continues to divide to form the **blastocyst**. The morula moves further into the uterus.
- The cells in the blastocyst are arranged into an outer **trophoblast** and an **inner cell mass**.
- The trophoblast gets attached to the uterine endometrium, and the process is called **implantation**. This leads to pregnancy.
- The inner cell mass gets differentiated to form the embryo.

Sex determination

- The sex of the baby would be decided during fertilisation.
- The chromosome pattern in the human female is XX and that in the male is XY.
- Therefore, all the haploid gametes produced by the female (ova) have the sex chromosome X.
- In male gametes (sperms) the sex chromosome could be either X or Y. Hence, 50% of sperms carry the X chromosome while the other 50% would carry the Y.
- After fertilisation the zygote would carry either XX or XY depending on whether the sperm carrying X or Y fertilized the ovum.
- The zygote carrying XX would develop into a female baby and XY would form a male
- Scientifically it is correct to say that the **sex of the baby is determined by the father and not by the mother.**

Pregnancy / Gestation

- After implantation, the trophoblast forms finger-like projections called **chorionic villi**, surrounded by the uterine tissue and maternal blood.
- The chorionic villi and the uterine tissue get integrated to form the **placenta**, which helps in supplying the developing embryo with oxygen and nutrients, and is also involved in the removal of wastes.

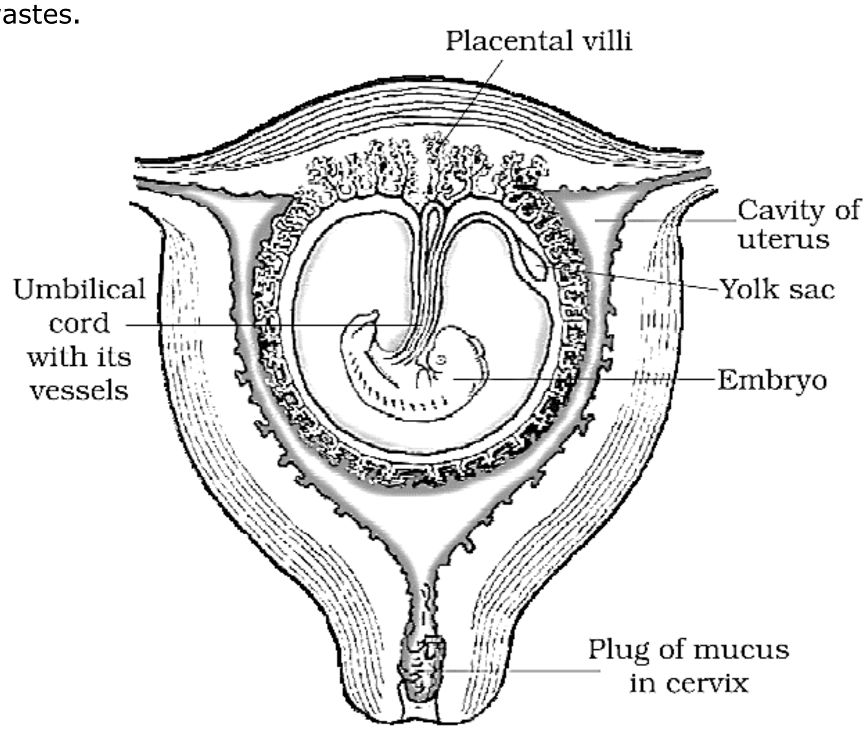

Figure: The human foetus within the uterus

- The placenta is connected to the embryo by the **umbilical cord**. The placenta acts as an endocrine gland, and produces the **human chorionic gonadotropins (hCG)**, **human placental lactogen (hPL)**, **oestrogen, progesterone** and **relaxin** (later stages of pregnancy).
- These hormones support foetal growth and help in the maintenance of pregnancy. Hormones like oestrogen, progestogen, cortisol, prolactin, etc., are increased several folds in the maternal blood.

Embryonic development

- Immediately after implantation, Blastocyst undergoes Gastrulation process.
- **Gastrulation -** the inner cell mass (embryo) gets differentiated into the three germ layers (embryonic layer) ectoderm, mesoderm and endoderm, which give rise to the different tissues. This ability of the inner cell mass is due to the presence of multi-potent cells called **stem cells**.
- After one month of pregnancy - the embryo's heart is formed.
- End of the second month of pregnancy - the foetus develops limbs and digits
- First trimester (by the end of 12 weeks) - most of the major organ systems are formed, for example, the limbs and external genetalia organs are well developed.
- Fifth month - the first movements of the foetus and appearance of hair on the head are usually observed.
- Second trimester (by the end of 24 weeks) - the body is covered with the fine hair, eye-lids separate and eyelashes are formed.
- At the end of the nine month of pregnancy - the foetus is fully developed and is ready for delivery.

Parturition and Lactation

- **Gestation period –** the duration of 9 months of human pregnancy is called the **gestation period**.
- **Parturition -** At the end of nine month, vigorous uterine contractions lead to the delivery of the foetus. This process is called **parturition**.
- Parturition is a **neuro-endocrine mechanism**, and is started by the signals from the fully developed foetus and the placenta, which induce mild uterine contractions called foetal ejection reflex.
- The mild uterine contraction caused by the signals from the fully developed foetus and the placenta is called **foetal ejection reflex**.
- Foetal ejection reflex causes the release of oxytocin from the posterior pituitary, which causes stronger uterine contractions.
- This leads to the expulsion of the baby along with the placenta.
- During pregnancy, the mammary glands undergo differentiation, and milk is produced during the end of pregnancy.
- **Colostrum -** The milk produced during the first few days of lactation is known as colostrum. It contains several antibodies that aid the newborn to develop resistance.

SUMMARY

Humans are sexually reproducing and viviparous. The male reproductive system is composed of a pair of testes, the male sex accessory ducts and the accessory glands and external genitalia. Each testis has about 250 compartments called testicular lobules, and each lobule contains one to three highly coiled seminiferous tubules. Each seminiferous tubule is lined inside by spermatogonia and Sertoli cells. The spermatogonia undergo meiotic divisions leading to sperm formation, while Sertoli cells provide nutrition to the dividing germ cells. The Leydig cells outside the seminiferous tubules, synthesise and secrete testicular hormones called androgens. The male external genitalia is called penis.

The female reproductive system consists of a pair of ovaries, a pair of oviducts, a uterus, a vagina, external genitalia, and a pair of mammary glands. The ovaries produce the female gamete (ovum) and some steroid hormones (ovarian hormones). Ovarian follicles in different stages of development are embedded in the stroma. The oviducts, uterus and vagina are female accessory ducts. The uterus has three layers namely perimetrium, myometrium and endometrium. The female external genitalia includes mons pubis, labia majora, labia minora, hymen and clitoris. The mammary glands are one of the female secondary sexual characteristics.

Spermatogenesis results in the formation of sperms that are transported by the male sex accessory ducts. A normal human sperm is composed of a head, neck, a middle piece and tail. The process of formation of mature female gametes is called oogenesis. The reproductive cycle of female primates is called menstrual cycle. Menstrual cycle starts only after attaining sexual maturation (puberty). During ovulation only one ovum is released per menstrual cycle. The cyclical changes in the ovary and the uterus during menstrual cycle are induced by changes in the levels of pituitary and ovarian hormones. After coitus, sperms are transported to the junction of the isthmus and ampulla, where the sperm fertilises the ovum leading to formation of a diploid zygote. The presence of X or Y chromosome in the sperm determines the sex of the embryo. The zygote undergoes repeated mitotic division to form a blastocyst, which is implanted in the uterus resulting in pregnancy. After nine months of pregnancy, the fully developed foetus is ready for delivery. The process of childbirth is called parturition which is induced by a complex neuroendocrine mechanism involving cortisol, estrogens and oxytocin. Mammary glands differentiate during pregnancy and secrete milk after child-birth. The new-born baby is fed milk by the mother (lactation) during the initial few months of growth.